

Tutorial 10 Solution

Solution 1

This is a summer with two inputs. Using Eq. (5.15),

$$v_o = - \left[\frac{10}{5}(2) + \frac{10}{2.5}(1) \right] = -(4 + 4) = -8 \text{ V}$$

The current i_o is the sum of the currents through the 10-k Ω and 2-k Ω resistors. Both of these resistors have voltage $v_o = -8 \text{ V}$ across them, since $v_a = v_b = 0$. Hence,

$$i_o = \frac{v_o - 0}{10} + \frac{v_o - 0}{2} \text{ mA} = -0.8 - 0.4 = -1.2 \text{ mA}$$

Solution 2

By voltage division,

$$v_{in+} = 25V \left(\frac{3\Omega}{5\Omega + 3\Omega} \right) = 9.375V$$

By the virtual short circuit between the input terminals, $v_{in-} = 9.375 \text{ V}$

Using **Ohm's** law, the current through the 15 Ω resistor is

$$I_{15} = \left(\frac{30V - 9.375V}{15\Omega} \right) = 1.375V$$

The input impedance is infinite; therefore, $I_{in-}=0$ and $I_{15}=I_{20}$.

Use Kirchoff's voltage law to find the output voltage at A.

$$v_A = v_{in-} - 20I_{20} = 9.375 \text{ V} - (20\Omega)(1.375 \text{ A}) = -18.125 \text{ V}$$

Solution 3

This op amp circuit is a summing amplifier. Since $i=0$,

$$i_f = \frac{v_1}{R_1} + \frac{v_2}{R_2} = \frac{3 \text{ V}}{8 \Omega} + \frac{2 \text{ V}}{4 \Omega} = 0.875 \text{ A}$$

$$v_o = -i_f R_f = -(0.875 \text{ A})(8 \Omega) = -7 \text{ V}$$

Solution 4

For OP-AMP -1: It is a inverting Op-Amp.

Therefore,

Apply KCL at inverting terminal

$$(0-V_i)/R_1+(0-V_{01})/R_2 = 0$$

$$V_{01}= (-R_2/R_1) V_i = -(1k/1k) \times 2 = -2 \text{ Volt} \quad \text{.....(i)}$$

For non-inverting amplifier, apply KCL again at inverting terminal for V_{out}

$$(V_{01}-0)/R+(V_{01}-V_{out})/R_f = 0$$

$$V_{out} = (1+R_f/R) V_{01}$$

$$V_{out} = (1+1/1) \times (-2) = -4 \text{ Volt}$$

Solution 5(a)

For Ideal case of Op-Amp, voltage gain (A_v)= $V_o/V_{id} = \infty$

Therefore, $V_{id} = 0$ i.e. $V_+ - V_- = 0$ or $V_+ = V_-$ (Virtual short concept)

Apply KCL at node A position in the inverting terminal

$$(0-V_i)/R_1+(0-V_o)/R_2 = 0$$

$$A_v = V_o/V_i = -R_2/R_1$$

Solution 5(b)

Apply superposition in the given Op-Amp

The output voltage (V_{01}) due to inverting input = $(-R_f/R)V_i = (-2k/1K) \times 1 = -2 \text{ Volt}$

and for $V_1 = 1k \times 1/1k+1k = 0.5$ (Potential divider)

The output voltage (V_{02}) due to non inverting input = $(1+R_f/R)V_1 = (1+2/1) \times 0.5 = 1.5 \text{ Volt}$

The total output (V_0) = $V_{01}+V_{02} = -2+1.5 = -0.5 \text{ Volt}$

Solution 6

By voltage division, $v_{in+} = v_i \left(\frac{2k\Omega}{3k\Omega} \right) = \frac{2}{3} v_i$

By the virtual short circuit, $v_{in-} = v_{in+} = \frac{2}{3} v_i$

$$i = \frac{v_{in-}}{3k\Omega} = \frac{\frac{2}{3} v_i}{3k\Omega}$$

Since the op amp draws no current, $i_f = i$

$$\frac{v_o - v_{in-}}{R_f} = \frac{\frac{2}{3} v_i}{3k\Omega}$$

But, $v_o = 5v_i$.

$$\frac{5v_i - \frac{2}{3} v_i}{R_f} = \frac{\frac{2}{3} v_i}{3k\Omega}$$

$$\frac{13}{3} = \frac{2}{3k\Omega}$$

$R_f = 19.5 \text{ k}\Omega$, Answer is B.